System Design Document (SDD)

System: Universal Absorbed Calculation System

1. What the System Is

The Universal Absorbed Calculation System is a web application for absorbed dose calculation.

It has two main parts:

1. Calculator page for doing calculations.
2. Admin portal for managing settings, data tables, formulas, and run history.

Users can open it in a browser without installing desktop software.

2. Who Uses It

1. Public users: can run calculations.
2. Admin/Physicist users: can manage datasets and formulas.
3. Viewer users: can view summary and run history.

3. What the System Does

1. Accepts beam and chamber input values from the user.
2. Gets temperature and pressure from approved sources:
   - Live location + weather service.
   - Approved dataset location.
3. Runs the dose calculation using the active formula and active datasets.
4. Shows the result clearly:
   - Dose per measurement.
   - Dose per 100 MU.
5. Stores a run record for traceability.

4. Why This System Design Was Chosen

This design was chosen because it supports the real day-to-day workflow:

1. Easy access: works in a web browser.
2. Easy adoption: users do not need to install special desktop tools.
3. Easy updates: admins can update datasets and formulas in the portal.
4. Better traceability: each run keeps inputs, outputs, and versions used.
5. Better governance: only validated datasets and approved formulas can be active.

5. How the System Builds Trust

The system maintains result quality in several ways:

1. Controlled data updates:
    - Uploaded datasets are checked before use.
    - Invalid datasets cannot be activated.
    - Only one active version is used at a time for each dataset type.
2. Controlled formula updates:
    - Formulas are validated before use.
    - Invalid formulas cannot be activated.
    - Only one active formula per beam type is used.
3. Run traceability:
    - Every run stores who ran it, when it ran, and what versions were used.
    - This supports audit and review.
4. Role-based access:
    - Sensitive actions are limited to authorized roles.
5. Password protection:
    - Login credentials are stored in a secure hashed form.

6. Simple End-User Guide

6.1 How to Run a Calculation

1. Open the calculator page.
2. Enter beam and chamber values.
3. Check location, temperature, and pressure shown at the top.
4. Click `Calculate Dose`.
5. Read the two output values.
6. Open details if you need intermediate values and context.

6.2 How to Read the Result

1. `Dose / measurement (Gy)` is the result for the entered measurement.
2. `Dose / 100 MU (Gy)` is normalized for 100 MU.

6.3 If Something Fails

1. Check if required datasets are active.
2. Check if a formula is active for the selected beam type.
3. Recheck unit selection and input values.

7. Simple Admin Guide

1. Set environmental source (Live or Dataset).
2. Upload and activate validated datasets.
3. Create, test, and activate formulas.
4. Review run history for audit and troubleshooting.

Recommended order:

1. Environment settings.
2. Datasets.
3. Formulas.
4. Run history review.

8. Data and Audit Records

The system keeps:

1. Users and roles.
2. Dataset versions and validation results.
3. Formula versions and validation results.
4. Calculation run history with full context.
5. App settings.

This supports transparency during internal or external review.

9. Security and Safety Summary

1. Users must sign in for admin actions.
2. Formula execution is restricted to safe math expressions.
3. Invalid data/formulas are blocked from activation.
4. Run history gives accountability for each calculation.

10. Known Boundaries

1. This system is for calculation workflow and governance, not enterprise identity management.
2. It depends on network availability for live weather/location updates.
3. Very high multi-user write traffic may require a stronger database setup in future.

11. System Summary

This system is designed with:

1. Simple user operation.
2. Controlled and validated scientific inputs.
3. Controlled and versioned formulas.
4. Clear role-based permissions.
5. Full run traceability for audit.

These controls make results explainable, reviewable, and repeatable.